[BUG] Error for user-defined device `dtype` with the TF interface

https://github.com/PennyLaneAI/pennylane/issues/2105

ROW 69

[BUG] Error for user-defined device `dtype` with the TF interface #2105

New issue

Closed #2120

antalszava opened on Jan 18, 2022

Expected behavior

No error arising related to `dtype` mismatch when setting custom `dtype` s on the device and using TF.

Actual behavior

The following error is raised with a QNode that takes floats as input and a TF loss:

```
InvalidArgumentError: cannot compute AddN as input #10(zero-based) was expected to be a float tensor but
```

Additional information

There might be a conversion happening in the backward pass (potentially in the custom gradient implementation). When omitting the use of `SparseCategoricalCrossentropy`, the gradient can be obtained without an issue.

Source code

```
import tensorflow as tf
import numpy as np
import pennylane as qml

nwires = 5
dev = qml.device("lightning.qubit", wires=nwires)
dev.C_DTYPE = np.complex64
dev.R_DTYPE = np.float32

@qml.qnode(dev, interface='tf', diff_method='adjoint')
def circuit(weights, features=np.zeros(nwires)):
    for i in range(nwires):
        qml.RX(features[i], wires=i)
        qml.RX(weights[i], wires=i)
    return [qml.expval(qml.PauliZ(0)), qml.expval(qml.PauliZ(1))]

np.random.seed(42)

ndata = 100
data = [np.random.randn(nwires).astype('float32') for _ in range(ndata)]
label = [np.random.choice([1, 0]).astype('int') for _ in range(ndata)]

loss = tf.losses.SparseCategoricalCrossentropy()

params = tf.Variable(np.random.randn(nwires).astype('float32'), trainable=True)
with tf.GradientTape() as tape:
    probs = [circuit(params, d) for d in data]
    loss_value = loss(label, probs)

grads = tape.gradient(loss_value, [params])
```

Tracebacks

No response

System information

Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Use `qml.math.cast` after `qml.math.convert_like` in `qml.gradients` to copy `dtype`

PennyLaneAI/pennylane

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Participants

```
Name: PennyLane
Version: 0.21.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /home/antal/xanadu/pennylane
Requires: numpy, scipy, networkx, retworkx, autograd, toml, appdirs, semantic_version, autoray, cachetools, pe
Required-by: PennyLane-Lightning, PennyLane-Cirq, PennyLane-Orquestra, pennylane-qulacs, amazon-braket-pennyla
Platform info:           Linux-5.11.0-44-generic-x86_64-with-glibc2.10
Python version:          3.8.5
Numpy version:           1.21.4
Scipy version:           1.7.3
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.20.1)
- cirq.mixedsimulator (PennyLane-Cirq-0.19.0)
- cirq.pasqal (PennyLane-Cirq-0.19.0)
- cirq.qsim (PennyLane-Cirq-0.19.0)
- cirq.qsimh (PennyLane-Cirq-0.19.0)
- cirq.simulator (PennyLane-Cirq-0.19.0)
- orquestra.forest (PennyLane-Orquestra-0.15.0)
- orquestra.ibmq (PennyLane-Orquestra-0.15.0)
- orquestra.qiskit (PennyLane-Orquestra-0.15.0)
- orquestra.qulacs (PennyLane-Orquestra-0.15.0)
- qulacs.simulator (pennylane-qulacs-0.17.0.dev0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.18.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.18.0)
- aqt.sim (PennyLane-AQT-0.18.0)
- projectq.classical (PennyLane-PQ-0.18.0.dev0)
- projectq.ibm (PennyLane-PQ-0.18.0.dev0)
- projectq.simulator (PennyLane-PQ-0.18.0.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.18.0.dev0)
- forest.qvm (PennyLane-Forest-0.18.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.18.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.19.0)
- ionq.qpu (PennyLane-IonQ-0.17.0.dev0)
- ionq.simulator (PennyLane-IonQ-0.17.0.dev0)
- strawberryfields.fock (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gaussian (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gbs (PennyLane-SF-0.20.0.dev0)
- strawberryfields.remote (PennyLane-SF-0.20.0.dev0)
- strawberryfields.tf (PennyLane-SF-0.20.0.dev0)
- default.gaussian (PennyLane-0.21.0.dev0)
- default.mixed (PennyLane-0.21.0.dev0)
- default.qubit (PennyLane-0.21.0.dev0)
- default.qubit.autograd (PennyLane-0.21.0.dev0)
- default.qubit.jax (PennyLane-0.21.0.dev0)
- default.qubit.tf (PennyLane-0.21.0.dev0)
- default.qubit.torch (PennyLane-0.21.0.dev0)
```

Existing GitHub issues

- [x] I have searched existing GitHub issues to make sure the issue does not already exist.

antalszava added bug on Jan 18, 2022

antalszava changed the title ~~[BUG] Error for customly set device `dtype` with the TF interface~~ [BUG] Error for user-defined device `dtype` with the TF interface on Jan 18, 2022

yitchen-tim on Jan 18, 2022

Adding info about the bug:
It only happens with big number of `ndata`. For example, there is no bug when `ndata=10` instead of 100. The value of `ndata` which starts triggering the bug also depends on the random seed. When the seed is 42, the bug happens when `ndata` is bigger than 14 on my machine.

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👍 2

josh146 on Jan 19, 2022 Member

Thanks @yitchen-tim for the details! Although I find it very odd that it is dependent on the size of the features 🤔

@antalszava is it like the case that there are more `float64` assumptions hardcoded in somewhere?

antalszava mentioned this on Jan 22, 2022

Use `qml.math.cast` after `qml.math.convert_like` in `qml.gradients` to copy `dtype` #2120

antalszava linked a pull request that will close this issue Use `qml.math.cast` after `qml.math.convert_like` in `qml.gradients` to copy `dtype` #2120 on Jan 24, 2022

antalszava closed this as completed in #2120 on Jan 25, 2022

Use `qml.math.cast` after `qml.math.convert_like` in `qml.gradients` to copy `dtype` #2120

Code

Merged antalszava merged 12 commits into `master` from `compute_vjp_use_cast_like` on Jan 25, 2022

Conversation 6 | Commits 12 | Checks 0 | Files changed 4 | +114 −5

antalszava commented on Jan 22, 2022 • edited — Contributor

Context:
`qml.math.cast_like` attempts to copy the dtype of a tensor whereas `qml.math.convert_like` only uses the type of the tensors during conversion

The `qml.gradients` module uses `qml.math.convert_like`. There may, however, be cases where the dtype would need to be copied, otherwise unexpected behaviour arises.

Description of the Change:

Benefits:

Possible Drawbacks:

Related GitHub Issues:
#2105

`use cast_like instead of convert_like in the gradient pipelines to pr…` … 8d84766

github-actions bot commented on Jan 22, 2022 — Contributor

Hello. You may have forgotten to update the changelog!
Please edit `doc/releases/changelog-dev.md` with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.

Reviewers
josh146
Jaybsoni

Assignees
No one assigned

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
[BUG] Error for user-defined device `dtype` ...

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3 participants